

Ethan Phan

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EXPERIENCE

Technical Support Programmer

Digital Provisions, LLC

February 2025 – Present

Ronkonkoma, NY (Remote)

- Maintain and enhance a proprietary physical security and project management platform used to oversee and support over 65,000 deployed security devices across K-12 schools, businesses, and public institutions.
- Collaborate with the CTO to identify and launch new features, including a one-click reporting generator, that improves technician field efficiency by 150%, directly enhancing service delivery.
- Employ knowledge of PHP, jQuery, and MySQL to update legacy code as well as identify and resolve application bugs and performance issues, improving overall system stability and user experience for 223 active users.

Research Assistant (Student Programmer)

Network Science Institute, Northeastern University

August 2024 – February 2025

Remote

- Collaborated with a research team investigating misinformation spread on Facebook, supporting NLP-driven analysis by engineering tools for large-scale data collection and pre-processing.
- Built a dynamic scraping pipeline using Python, Selenium, and Google API to extract post-level data from 100+ Facebook profiles, enabling scalable research and analysis of over 10,000 posts with minimal manual intervention.

Teaching Assistant

School of Engineering and Applied Sciences, University at Buffalo

August 2022 – December 2024

Buffalo, NY

- Taught core CS concepts such as Data Structures, Societal Computing, and Discrete Mathematics to 600+ students through interactive lectures and recitation sessions.
- Led four 1-hour recitations, addressed over 50 student questions, and held 5+ hours of office hours each week.
- Proposed areas for course improvement and provided actionable recommendations to enhance structure and engagement during weekly course staff meetings.

Technology Intern

M&T Bank

June 2023 – August 2023

Buffalo, NY

- Integrated Docker into GitLab CI/CD pipelines to automate deployment for 200+ .NET web applications, reducing deployment time and manual errors.
- Directed the design of deployment architecture, enhancing performance and sustainability, resulting in \$500,000 in annual savings for the bank.
- Developed a full-stack web application and CI/CD pipeline using Angular, C#, .NET, and YAML in an Agile team, enhancing cross-team productivity and organization.

PROJECTS

Convolutional Neural Network | *Python, PyTorch, NumPy*

- Constructed a neural network in PyTorch to classify MNIST handwritten digit images, achieving 98.27% accuracy.
- Enhanced model performance using Stochastic Gradient Descent (SGD) and compared results across batch sizes to evaluate convergence speed, performance and training efficiency.

Business Scheduling Platform | *React, Tailwind CSS, PHP, MySQL*

- Developed and deployed a full-stack business scheduling application with separate administrative and client UI, promoting organization and accountability for small businesses and their workflow.
- Conducted load testing with 100+ concurrent users to ensure application stability.
- Devised project design and directed front-end development within a Scrum team of 5, managing tasks on Trello.

EDUCATION

University at Buffalo, The State University of New York

Bachelor of Science in Computer Science

August 2021 – February 2025

GPA: 4.000

- Honors: Pride of New York Scholarship, Dean's List

TECHNICAL SKILLS

Languages: Python, PHP, C/C++, C#, Java, JavaScript, HTML/CSS

Web Development: Angular, React, SCSS, Tailwind CSS, TypeScript, jQuery, Express, .NET, Flask

Technologies & Tools: UNIX, Git, GitHub, GitLab, Node.js, MySQL, MongoDB, Docker

Product Management Tools: Trello, Jira, Confluence